higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATES (VOCATIONAL)

ENGINEERING PROCESSES
NQF LEVEL 4

(6021004)

8 March 2019 (Y-Paper)
13:00–16:00

This question paper consists of 5 pages.

NCV

TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

NCV

QUESTION 1

1.1 It can be dangerous to work in a workshop if the safety precautions are not followed.

State FOUR important safety precautions that must be taken into consideration when working in a workshop. (4)

1.2 Explain why is it necessary for an artisan to put a lockout system in place before maintenance work can commence. (2)

[6]

QUESTION 2

2.1 As an apprentice you are requested to replace machine parts and assemble them according to the manufacturer's manual or the job card.

State FOUR important factors that you will need to consider when performing this task. (4 × 2) (8)

2.2 Choose a description from COLUMN B that matches an item/a word in COLUMN A. Write only the letter (A–G) next to the question number (2.2.1–2.2.6) in the ANSWER BOOK.

COLUMN A		COLUMN B	
2.2.1	Metallic surface	A	pieces of non-metallic material that are in the mould and become part of the cast when it solidifies
2.2.2	Cavities in the cast		
2.2.3	Discontinuities	B	too little material was used when the metal was cast into the mould
2.2.4	Defective surfaces	C	occurs when the metal does not shrink properly
2.2.5	Incomplete casting		
2.2.6	Inclusions	D	tiny, rough or smooth holes that are on the surface of the cast and therefore the surface is not smooth
		E	cracks and tears in the cast
		F	pieces of metal that protrudes from the cast
		G	surfaces with folds or scars or sand stuck to them

(6 × 1) (6)

2.3 State THREE reasons for pre-cleaning a component before dismantling it. (3 × 2) (6)

2.4 Name SIX hand tools that can be used when assembling an engine of a vehicle. (6)
[26]

QUESTION 3

3.1 Workers and students using machines and power tools are faced with a potential risk of injury due to the fast-moving parts of these machines.

State FOUR safety precautionary measures to be observed when operating a pedestal drill. (4)

3.2 In engineering there are various ways for checking tolerances on machine components.

Briefly explain THREE methods to check for tolerances on machine components. (3)

3.3 State FOUR reasons for cleaning the CNC machine after use. (4)
[11]

QUESTION 4

4.1 State the SIX steps that need to be followed when preparing for oxyacetylene cutting. (6)

4.2 Certain band saws make use of a mist coolant system.

Briefly describe how this system operates in order to cool the workpiece and the band. (4 × 2) (8)

4.3 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (4.3.1–4.3.6) in the ANSWER BOOK.

4.3.1 When selecting cutting blades for a machine you don't have to consider the size, shape and the thickness of the materials.

4.3.2 Horizontal band saws can be used for cutting different shapes.

4.3.3 Regulators are valves that are used to control the flow of gases.

4.3.4 An impact is when the force of one object effects the positon another.

4.3.5 An alloy is a compound of two metals that have been mixed together.

4.3.6 Mechanical engineers are involved with the building of structural works.

(6 × 1) (6)

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4.4 When you are going to cut material with power cutting machines, you must understand the type of material and its composition, as this will determine the cutting process and the selection of the correct equipment.

State SIX general guidelines that you should follow when setting up a cutting machine for mechanical cutting. (6)
[26]

QUESTION 5

5.1 Before you start to produce components using a CNC machine, you have to simulate the process on the computer or work station screen to identify errors.

State FOUR factors which need to be monitored before adjustments can be made to the program. (4 × 2) (8)

5.2 Explain how X-, Y- and Z-coordinates function when you are operating a CNC machine. (3)

5.3 State THREE advantages of using a CNC machine compare to a normal centre lathe. (3)

5.4 It is important for an artisan to monitor the path of the cutting tool because if the tool strays from the correct path, the process can be stopped.

State THREE aspects that will be used to monitor the cutting process. (3)
[17]

QUESTION 6

6.1 State FOUR steps that you need to follow to ensure your safety when lighting up or shutting down gas welding equipment. (4)

6.2 When a metal corrodes easily, it can be coated with another metal that does not corrode easily.

State FIVE methods that can be used to coat the metal. (5)

6.3 The joining process is a simple technique of joining two solid objects to form an assembly.

State FIVE such joining methods that are used in industry. (5)
[14]

TOTAL: 100

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